

Part Two: Take-Home Examination

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Moderator: Mr RQ Odendaal

Release Date: 13 June 2020 at 12:00
Due Date: 13 June 2020 at 17:00

Please read and understand the following.

1. All test and examination regulations of the University of Pretoria apply.
2. This is an OPEN BOOK assignment. You may use any source available to you, like your textbook and lecture notes, to assist in answering the questions. However, you **MAY NOT** receive (or provide) help in any way whatsoever from (or to) any other person. This includes the lecturer, tutor, a friend and family members.
3. Answer the questions **in your own handwriting** on separate paper, lined or unlined.
Take quality photographs of your answer sheets and scan them as a PDF document. Use apps like **CamScanner** to convert the photographs to PDF.
Submit your answer sheets, via ClickUP, in the “FSK116 June Exam 2020” folder under the “ONLINE” headline. **We urge you to submit before 17:00.**
There will be a queue during submission. You may not get your submission receipt immediately! To accommodate this, we give you a grace period of 30 minutes only between 17:00 and 17:30 to make your submission.
YOU HAVE BEEN WARNED: Late submissions will not be accepted!
4. Number and label your submissions as in the question paper. Incorrectly-numbered and incorrectly-labelled answers may not be marked.
5. Provide neatly labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct numbers of significant figures, units, etc. Failure to do so will result in marks being deducted.
6. Your answer needs to be written in a systematic and logical fashion. Map out the steps you will take to reach your goal i.e. solve the problem in your mind first! You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not necessarily mean that you will receive full marks!
7. Units must be included when numbers are substituted; else, marks will be deducted.
8. You will need to look up constants and typical values for parameters in your calculations.
9. If any misconduct with respect to this examination is suspected, you will receive a mark of zero and will have to attend a formal disciplinary hearing, which may result in your expulsion from the University of Pretoria.

DECLARATION OF ORIGINALITY

UNIVERSITY OF PRETORIA - Department of Physics

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I understand what plagiarism is and am aware of the University's policy in this regard. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

This statement is a reminder of the Code of Conduct that you have already signed.

TOTAL POINTS: 43

QUESTION 1 [13]

Suppose that due to a gravitational torque exerted by the Moon on the Earth, our planet's rotation slows at a rate of 1.30 ms/century (see picture below). Show the full calculations, including the conversions. Do not round up in intermediate calculations.

HINT: You will need to use basic definitions of: α , $\Delta\omega$, $\Delta\theta$, and Δt to crack the problem.



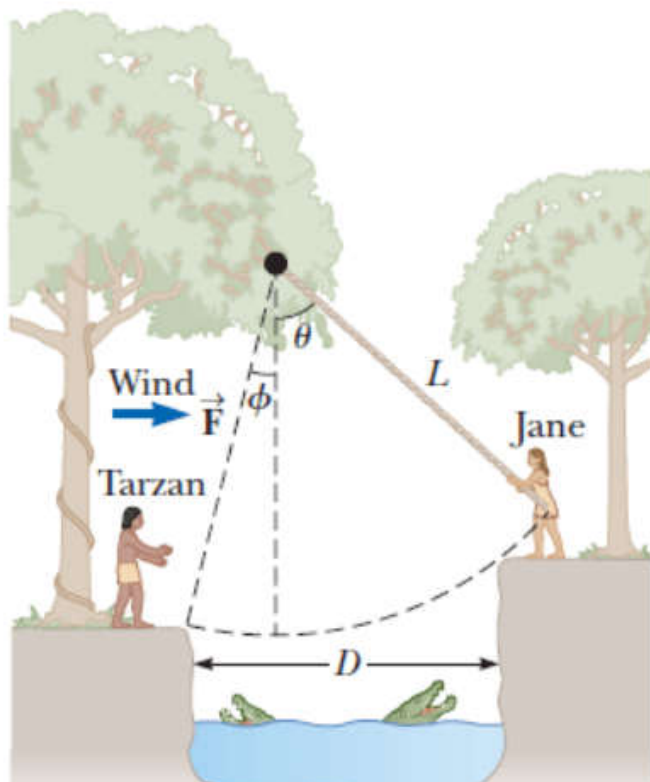
- (a) Calculate the Earth's angular acceleration due to this effect. (8)
- (b) Calculate the torque exerted by the Moon on the Earth. (3)
- (c) Calculate the length of the wrench / spanner an ordinary person would need to exert such a torque, as in the figure above. Assume the person can brace his feet against a solid firmament and exert a 850 N force. (2)

QUESTION 2 [14]

Study the figure below. In it, Jane and Tarzan stand on opposite banks of a river of width D . Jane's bank is higher than Tarzan's. Jane holds a rope of length L , the other end of which is attached to a tree over the river, making an angle of θ with the vertical. A dotted line shows that if the rope were to be held at Tarzan's position rather than Jane's, it would make an angle of ϕ with the vertical. A horizontal arrow labelled vector $\mathbf{F}$ shows that wind acts away from Tarzan and toward Jane. Utilize the following parameters in your calculation: $D = 51.0$ m, $L = 40.0$ m, $\theta = 50.0^\circ$, Jane's mass = 50.0 kg and $F = 111$ N.

HINTS: You may assume that the sector that subtends angle $(\phi + \theta)$ is approximately equal to D , the width of the river. Utilize the relation: $D = (L \sin \phi + L \sin \theta)$ to find ϕ .

You need to anchor / zero the gravitational potential energy at a common point e.g. where the rope attaches to the tree. In addition, remember to include the wind in your calculations, as an agency doing work.



(a) With what minimum speed (in m/s) must Jane begin her swing to just make it to the other side? Show all your steps, assumptions and the reasoning behind them. (10)

(b) Shortly after Jane's arrival, Tarzan and Jane decide to swing back across the river simultaneously).

With what minimum speed (in m/s) must they begin their swing? Assume that Tarzan has a mass of 80.0 kg.

(4)

QUESTION 3 [10]

With this problem, we want to explore the idea that, if it were not for drag, raindrops would attain fantastic speeds near Earth's surface. Piet observes that the raindrops that are hitting him, have a radius of 2.00 mm and fall from a cloud located 1000 m above the ground he is laying on. Take the drag coefficient of the raindrops to be 0.50 and the ambient temperature to be 20.0 °C.

HINT: The area in the terminal velocity formula is a cross-sectional area of the object under consideration.

(a) Determine the terminal speed of the raindrops. You may assume spherical raindrops made out of water only.

(4)

(b) Piet notices that the expression for the terminal speed used above does not contain explicit reference to the height of the cloud from whence the raindrops originate. Would the terminal speed for the raindrop in (a) be equal to that for a raindrop released 1.00 m above ground? Is there a discrepancy? Give reasons?

(3)

(c) Determine the final velocity of the raindrop (in km/h) in when we ignore air resistance,

(3)

QUESTION 4 **[6]**

A steel calorimeter cup with a specific heat capacity of $490 \text{ J/(kg}\cdot^{\circ}\text{C)}$ contains kerosene with a specific heat capacity of $2\,018 \text{ J/(kg}\cdot^{\circ}\text{C)}$ at 16.0°C . The combination is heated uniformly so that the temperature changes by 2.10°C per minute. If the cup has a mass of 300 g and contains 900 g of the liquid kerosene, determine the rate at which heat energy is added (in W).

(6)

HINTS: You need to specify what your system is.

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